

Transcriptomic Landscape of Cellular Senescence

Analysis across multiple senescence models (IR, RE, Dox, HRAS)

Abstract

This report presents a comprehensive RNA-seq analysis of various cellular senescence models, including Ionizing Radiation (IR), Replicative Exhaustion (RE), Doxorubicin (Dox), and HRAS-induced senescence. The study aims to identify a core transcriptional signature common across these different triggers. Our findings highlight significant differential expression of markers associated with cell cycle arrest, inflammatory response (SASP), and chromatin remodeling. Integration of multi-model data reveals a set of consistently upregulated genes (e.g., CCND3, ARRDC4) and downregulated genes (e.g., CDKN2C), providing insights into the convergent pathways of cellular senescence.

Introduction

Cellular senescence is a state of permanent cell cycle arrest characterized by morphological changes, the secretion of pro-inflammatory factors (SASP), and metabolic shifts. While triggered by various stressors, it is hypothesized that these models share a common molecular core. This analysis utilizes bulk RNA-seq to characterize the transcriptomic shifts across different senescence induction methods in human cells.

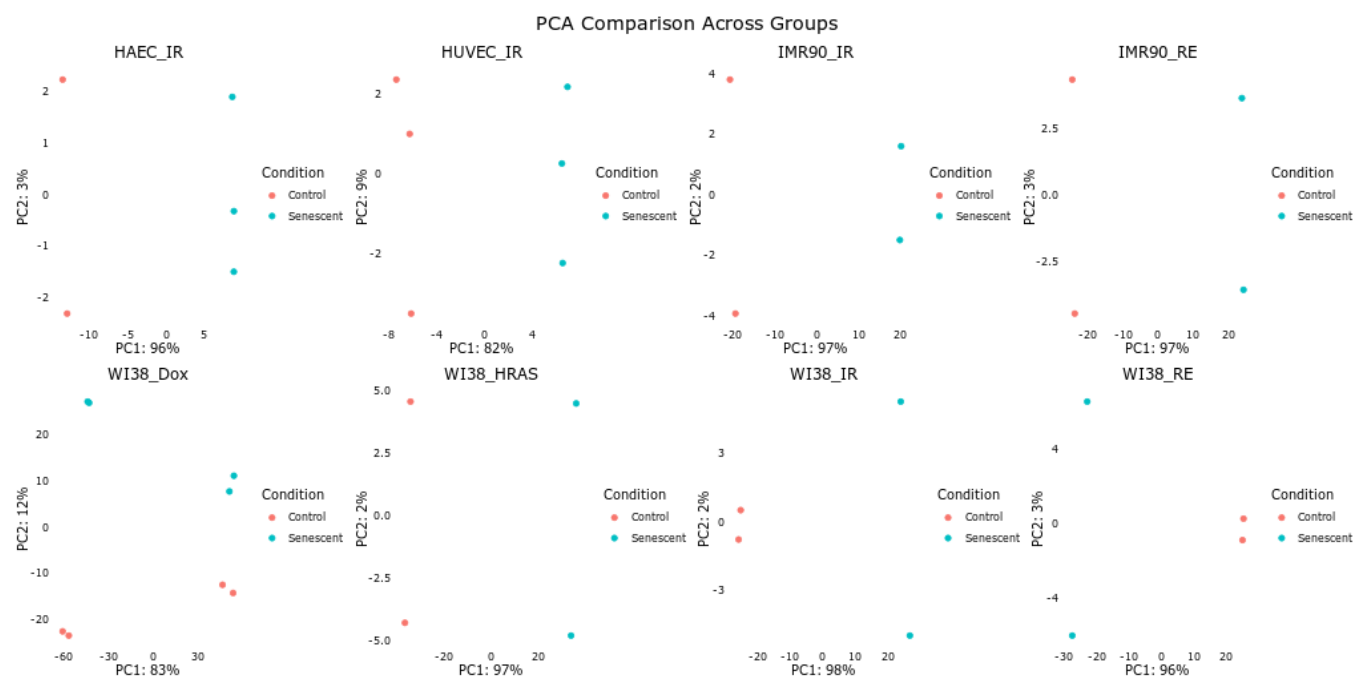
Methods

RNA-seq data was processed using a standard pipeline. Read counts were normalized and differential expression was performed using DESeq2. The experimental design accounted for biological replicates within each model. Statistical significance was defined by adjusted p-value < 0.05 . Principal Component Analysis (PCA) was used for dimensionality reduction. Functional enrichment analysis was conducted using GSEA and GO terms via clusterProfiler.

Results

The quality control analysis showed consistent library sizes and high correlation between replicates.

Figure 1: Principal Component Analysis (PCA)



PCA demonstrates a clear separation between control and senescent samples across all models, with the primary axis of variation corresponding to the senescence status.

Differential Expression Analysis

A core set of genes was identified as consistently differentially expressed across models. We found 5 core upregulated and 12 core downregulated genes.

Table 1: Core Upregulated Genes

X
CCND3
ARRDC4
PAM
GPR155
TNFRSF10C
PTCHD4

Figure 2: Representative Volcano Plot (HAEC IR)

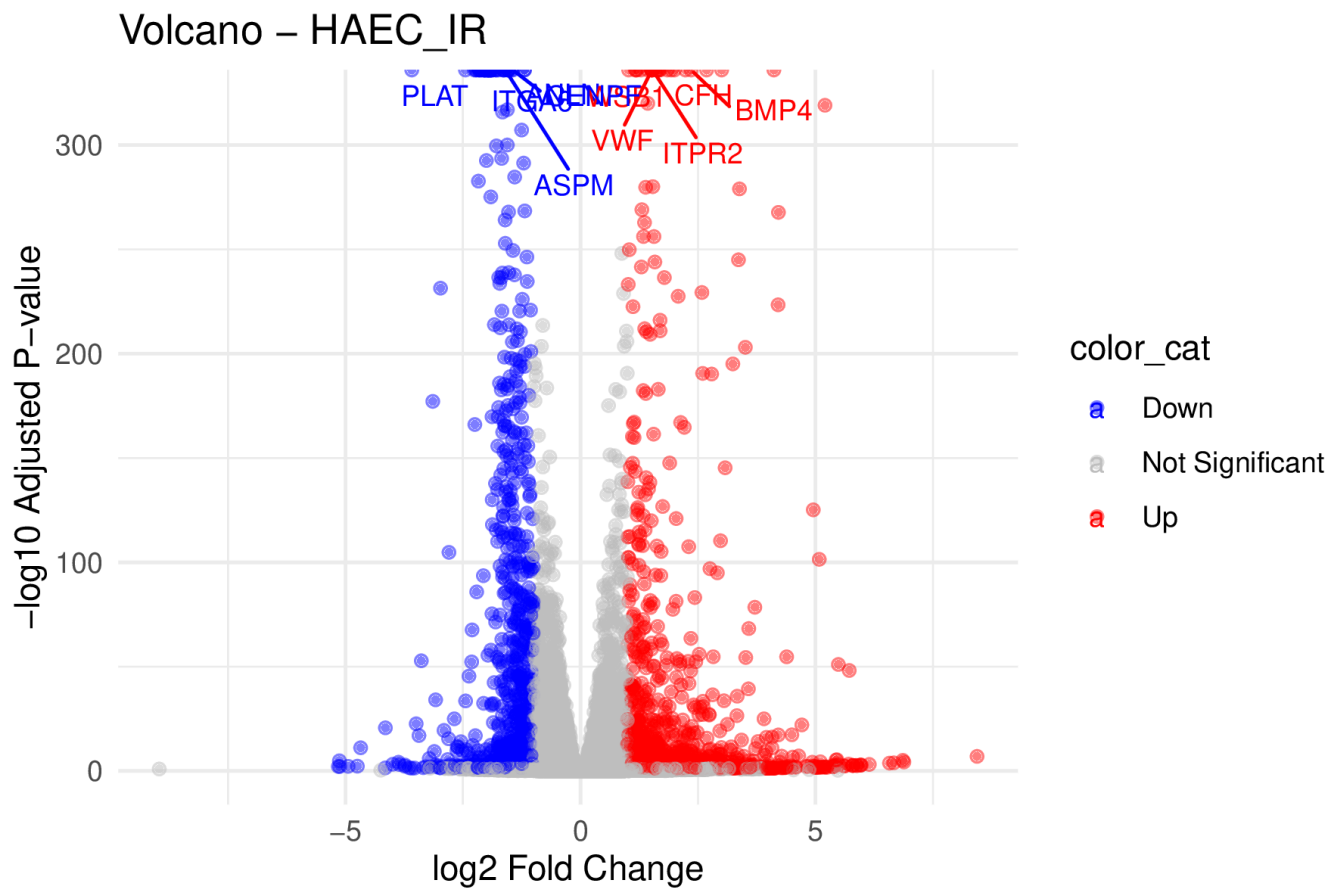
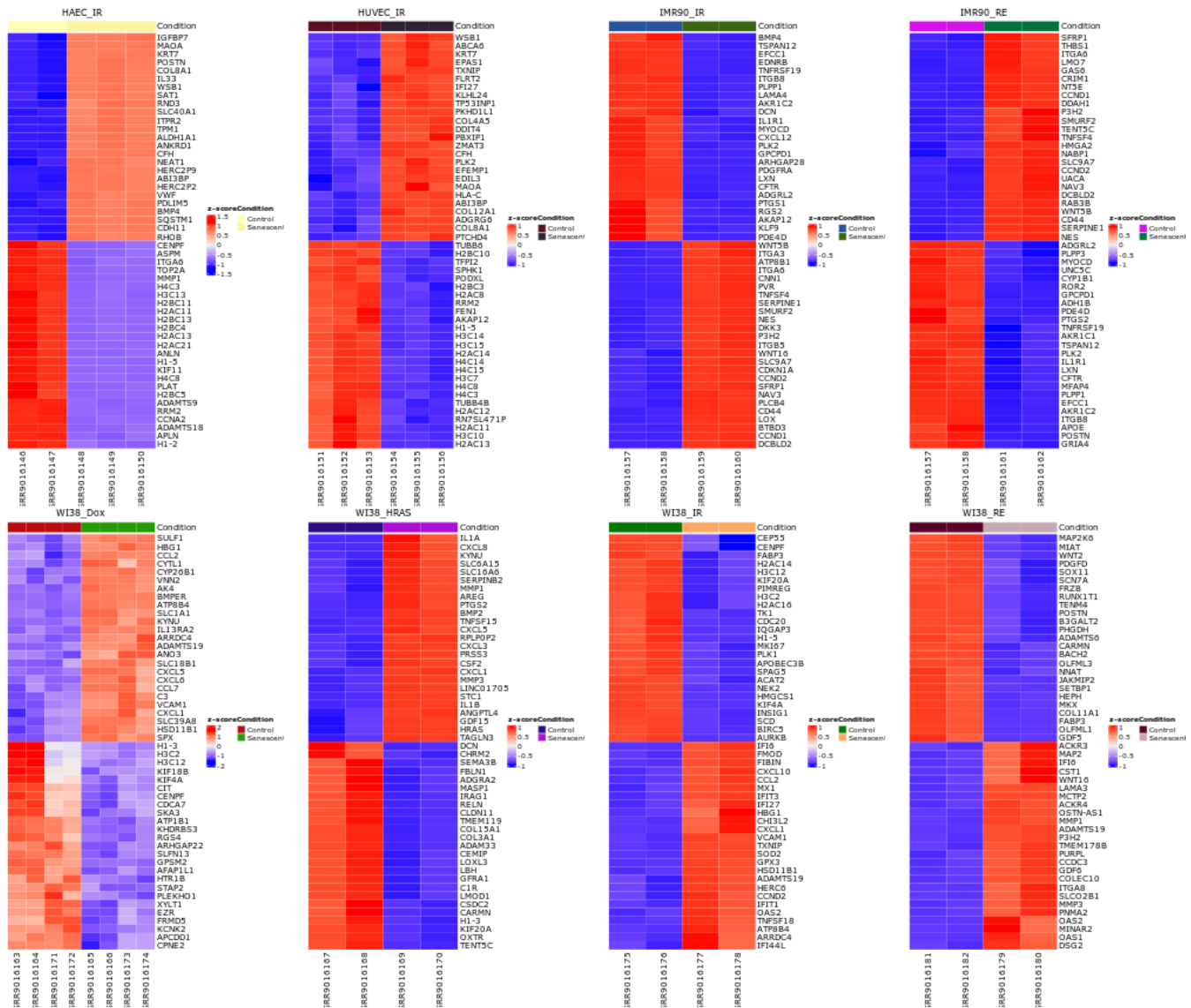


Figure 3: Core Differential Expression Heatmap



Functional Enrichment

GSEA revealed a significant downregulation of cell cycle-related processes, specifically those involved in chromosome segregation and DNA replication.

Table 2: Top Enriched GO Terms (Downregulated)

Description	NES	p.adjust
sister chromatid segregation	-2.73464282995773	5.51727272727273e-09
mitotic sister chromatid segregation	-2.72295800290955	5.51727272727273e-09
chromosome segregation	-2.68856734562519	5.51727272727273e-09
nuclear chromosome segregation	-2.65906269823136	5.51727272727273e-09
mitotic nuclear division	-2.64964102438843	5.51727272727273e-09
regulation of chromosome organization	-2.62482897262848	5.51727272727273e-09
DNA replication	-2.60983132109212	5.51727272727273e-09
DNA-templated DNA replication	-2.58578142969589	5.51727272727273e-09

nuclear division	-2.58348525125622	5.51727272727273e-09
protein-DNA complex assembly	-2.5825942805666	5.51727272727273e-09
protein-DNA complex organization	-2.57518900564917	5.51727272727273e-09
microtubule cytoskeleton organization involved in cell cycle	-2.56886671014476	5.51727272727273e-09
organelle fission	-2.54344276056537	5.51727272727273e-09
spindle organization	-2.54097004853886	5.51727272727273e-09
mitotic spindle organization	-2.53673316811719	5.51727272727273e-09

Discussion

The consistency of the core gene set across different senescence triggers supports the existence of a universal senescent program. The downregulation of mitotic proteins and upregulation of SASP-related factors are hallmarks that were strongly observed in our data.

Conclusion

In conclusion, this study provides a high-resolution transcriptional map of senescence and identifies key genes that define the senescent state regardless of the inducing stimulus.

Figure Legends

Figure 1: PCA plot showing sample distribution. Colors represent different models; shapes represent control vs senescent.

Figure 2: Volcano plot for HAEC IR model. Points signify genes with adj p-value < 0.05 and $|\log_2FC| > 1$.

Figure 3: Hierarchical clustering heatmap of the core senescence signature.